

ECE 4240 Midterm Project Report: LiDAR-Based ...

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1 Modeling and Simulating a Virtual LiDAR

The LiDAR (Light Detection and Ranging) is an active sensor that provides accurate depth and reflection strength measurements via continuous firing of laser beams. The density of depth and intensity data that a LiDAR can capture within a full revolution is dependent on its hardware configurations. Specifically, the LiDAR's channels define the specific angles and amount of laser beams fired within one instance, while the LiDAR's rpm defines the maximum amount of times that lasers are fired within a full revolution. In this paper, LiDAR channels are synonymous with vertical θ angle resolution and the set of vertical angles in which lasers are fired, while rpm is more geometrically interpretable if described as horizontal ϕ angle resolution. Then, it is very feasible to simulate the beam-pattern of LiDARs via a ray-caster model, especially if we define the LiDAR's body frame $\mathcal{F}_{\mathcal{L}}$ to be in spherical coordinates:

1. Define $\phi = 0$ and $\theta \in [\theta_{min}, \theta_{max}]$ as the initial angular sector formed by the set of laser beams fired at $t = 0$.
2. Each laser beam is represented as a ray with its fired vertical angle θ , horizontal angle ϕ and range r . In other words, given an environment with obstacles and surfaces, for each channel and azimuth, a ray is cast from origin of the LiDAR until point of collision or sensor maximum range.
3. Block sizes represent the total number of times a LiDAR fires before outputting the set of all corresponding rays. For every block outputted, we convert the associated rays into cartesian points and range graph elements.

2 Representation of Scans as a Range Graph

A range graph is a 2D array representation of a LiDAR scan, specifically encoding vertical and horizontal angle along with cartesian (x, y, z) information of each laser beam in an ordered structure that models the LiDAR's hardware configurations. Range graph elements are represented as $\text{RangeGraph}[\theta][\phi]$, which contains 3D point of collision (x, y, z) , range r , and reflection intensity of a laser fired at vertical angle θ and horizontal angle ϕ .

3 Two-Layer-Graph Clustering

Point cloud segmentation allows us to recreate obstacle regions and geometry through exploitation of LiDAR hardware configuration and the physical nature of objects in 3D space. Obstacle regions

can be used for tracking algorithms that better inform longer-horizon planning, along with point cloud compression and cohesive visualization of the environment around the sensor platform.

”Two-Layer-Graph Clustering for Real-Time 3D LiDAR Point Cloud Segmentation” is a paper written by Yang, Wang, Lin, et. al on constructing range graphs given prior knowledge of a LiDAR’s beam firing patterns, then performing two separate segmentations on given range graph for accurate and real-time clustering.

First segmentation. Clustering horizontally adjacent points to populate a set graph. This serves to significantly reduce the amount of points that need to be traversed and explored in the second segmentation by first conducting a parallelizable clustering among points within the same channel or vertical angle.

Second segmentation. Merging vertically adjacent nodes via BFS or possible other graph traversal methods over a reduced set of set nodes. Given a SetNode, we search for vertically adjacent nodes that intersect with the current horizontal segment, as defined by RangeGraph node’s $(\theta_{start}, \theta_{end})$. Specifically,

4 KLT Tracking via Intensity-Depth Fusion

Proposal. The idea here is to simulate a form of camera-LiDAR sensor fusion without actually requiring synchronization between a physical camera and LiDAR. The RoboSense Helios LiDAR outputs UDP packets containing not only vertical θ and horizontal ϕ firing angles of each laser, the range r travelled prior to collision, and associated timestamps t , but also the reflection strength of each laser, also known as intensity. Reflection strength is dependent on the material of the surface that each laser beam collided with. As a result, we can essentially construct a sparse image of the environment around the sensor platform via an intensity map. This intensity map can be passed in with associated depth information to the KLT algorithm to perform some pseudo-optical flow on a sequence of LiDAR scans. ”The intensity measured by the LiDAR can be decomposed into four main factors under the Lambertian reflectance assumption: measurement distance, the surface reflectance, the strength of the incident ray, and the incident angle” (Dai et al.)

The relationship intensity, distance to target or range, and the angle at which the signal strikes the target (angle of incidence) is encapsulated by the LiDAR intensity equation:

$$I(R, \alpha, \rho) \propto P_r(R, \alpha, \rho) = \eta(R) \frac{I_e \rho \cos(\alpha)}{R^2},$$

where I_e denotes the laser’s emission power, R signifies the distance to the surface, α represents angle of incidence, and ρ is the parameter indicating the target’s reflectivity. The phenomenon of lens defocusing in LiDAR sensors results in a near-range effect $\eta(R)$ that impacts the intensity data at closer ranges ($R_n < 12m$). This causes the intensity equation to be inapplicable within these proximities, and as a result, we will adopt a modeling approach for the near-range effect:

$$\eta(R) = 1 - \exp \frac{-2r_d^2(R+d)^2}{D^2 S^2}$$

with r_d denoting the radius of the laser detector, d representing the offset between the measured range and object range, D is the diameter of the lens, and S signifies focal length, respectively.

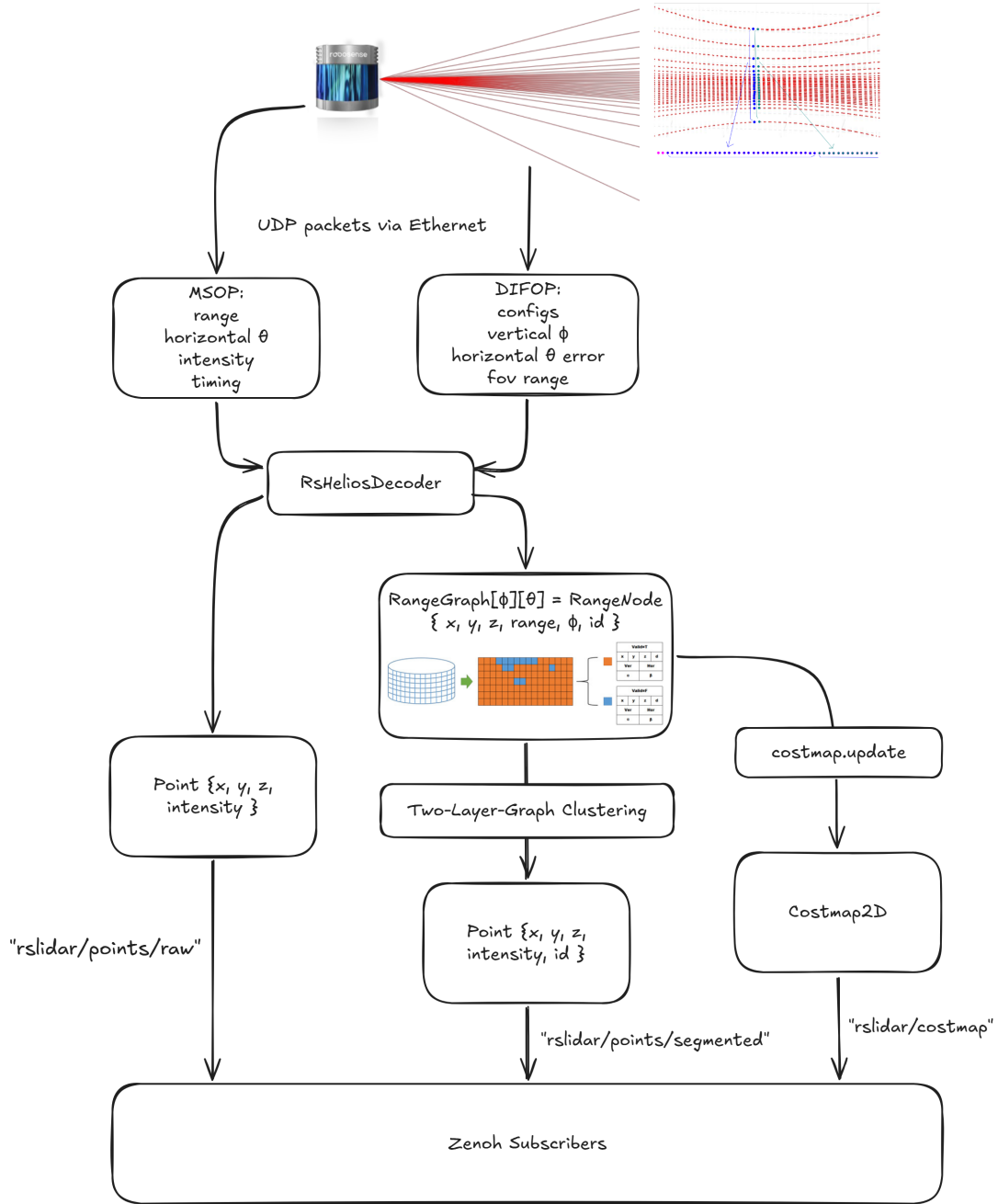


Figure 1: Training PL histogram with equal-frequency (left) and equal-width (right) bin edges, $K = 20$.